MIL-T-21578A
28 January 1966
Superseding
MIL-T-21578(Aer)
11 August 1958

MILITARY SPECIFICATION

TEST EQUIPMENT; HAZARDOUS LOCATION, INSTALLED; BASIC REQUIREMENTS FOR

This specifications mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers installed hazardous location test equipment for aircraft, engine, guided missiles and related accessories and components thereof and shall become a part of a detailed specification by attachment or reference. Hazardous location equipment is defined as test equipment operating in a hazardous location due to explosive vapors or utilizing test fluids having a flash point of 170° F or less, or when test conditions exceed the flash point of the test fluid,

2. APPLICABLE DOCUMENTS

2.1 Government documents normally furnished - The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)
QQ-Z-325	Zinc Coating, Electrodeposited, Requirements for
TT-C-490	Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings
TT-P-636	Primer Coating, Synthetic, Wood and Ferrous Metal

FSC 4920

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:57Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-T-704	Treatment and Painting of Material
MIL-M-3171	Magnesium Alloy, Process for Pre-treatment and Prevention of Corrosion on
MIL-F-3541	Fittings, Lubrication
MIL-S-5002	Surface Treatments and Metallic Coatings for Metal Surfaces of Weapon Systems
MIL-A-5498	Accumulators, Aircraft Hydropneumatic Pressure
MIL-F-5509	Fittings, Flared Tube, Fluid Connection
MIL-C-5541	Chemical Films and Chemical Film Materials for Aluminum and Aluminum Alloys
MIL-F-7024	Fluids, Calibrating, for Aircraft Fuel System Components
MIL-P-7105	Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT, General Requirements for
MIL-P-8585	Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity
MIL-F-18280	Fittings, Flareless Tube, Fluid Connection
MIL-P-26915	Primer Coating, Zinc Dust Pigmented, for Steel Surfaces

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:57Z
Check the source to verify that this is the current version before use.

STANDARDS

Military (Continued)

MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MS-28889	Valve, Air, High-Pressure Charging
MS-33586	Metals, Definition of Dissimilar

Air Force Navy Aeronautical Design

AND10050	Bosses, Standard Dimensions for Gasket Seal Straight Thread
AND 20001	Drive, Type XI Engine Accessory
AND 20002	Drive, Type XI Engine Accessory

(When requesting applicable documents ,refer to both title and number. Copies of unclassified documents may be obtained from the Commanding Officer, Naval Supply Depot (Code 1051), 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120. Requests for copies of classified documents should be addressed to the Naval Supply Depot, via the cognizant Government inspector.)

2.2 Other publications - The following documents form apart of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Board of Fire Underwriters Standards

Article 250	National Electrical Code
Article 500	National Electrical Code

(Copies of National Electrical Code Standards may be obtained from the National Board of Fire Underwriters, 85 John Street, New York 38, N. Y.)

3. REQUIREMENTS

3.1 Preproduction - The test equipment furnished under this specification shall be a product which has been inspected and passed the pre-production inspection specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:57Z
Check the source to verify that this is the current version before use.

3.2 Selection of specifications and standards - Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials - Materials shall conform to applicable specifications and shall be as specified herein and on applicable drawings. Materials which are not covered by specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

3.3.1 Metal parts - All metal parts shall be of a corrosion resistant material or treated in a manner to render them adequately resistant to corrosion (see 3.6).

.3.3.1.1 Dissimilar metals - Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

3.4 Design and construction - The design and construction of the test stand, various test stand systems, and components shall be in accordance with the following subparagraphs.

3.4.1 Grouping of systems, component, and cabinet - Design, layout, and assembly of the cabinet and its component parts shall be such as to facilitate quantity production and ease of maintenance. The test equipment shall be so designed that no special tools are! required for adjustment, cleaning, or maintenance of the test equipment or any component.

3.4.1.1 Structure - All test equipment shall be located within a fully enclosed metal cabinet suitably designed to withstand all loads imposed during service use and shipment. The cabinet shall provide suitable guards or shields for fan blades, heating coils, electrical connections, and all other component parts which may present a hazard to the operator. Instrument panels shall be constructed of not less than 11 gage metal; doors and cabinet panels shall be not less than 14 gage metal.

3.4.1.1.1 Size - Cabinet size shall not exceed 10 feet in length by 8 feet in depth (depth from operator’s side of stand to rear) by 7-1/2 feet high. Test equipment requiring a cabinet of larger size shall be constructed in sections to meet the size requirements. All instrumentation located above 6 feet high shall be inserted in the cabinet so as to be readily visible and accessible to the operator.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:57Z
Check the source to verify that this is the current version before use.

3.4.1.1.1.1 Sections - Equipment that must reconstructed infections to meet the size requirements shall be so designed that disassembly for shipment and subsequent re-assembly shall not create an alignment problem of the components affected. All mating parts, fluid and pneumatic lines, and electrical connections shall be properly identified by permanent tags and name plates to insure proper re-assembly.

3.4.1.1.2 Base design - All cabinets and sections shall include longitudinal channels which shall permit positioning or moving by means of a fork-lift truck. The base shall be a permanent part of the unit, of sufficient strength and rigidity to support the load without undue stress or strain on the test equipment, and shall be designed to distribute the floor load evenly over the floor space occupied by the test equipment, Channels (or other design as applicable) shall be a minimum of 8 inches in width and spaced a maximum of 20 inches between centers. Charnels shall be a minimum of 2 inches deep and shall be located to facilitate assembly of test equipment built in sections. Channels shall be located as near the center of gravity of the test equipment as practicable.

3.4.1.1.3 Accessibility and serviceability - Components shall be arranged within the cabinet to facilitate easy removal, maintenance, and servicing. Lubrication fittings, test locations, filters, shaft seals, and similar components shall be accessible without disassembly or removal of other components. All other components shall be readily maintainable. All internal components shall be accessible through rear and side removable doors having quick-detach type hinges and provided with positive flush-type or streamlined hand latches which are easily operated. The hinge pins shall be tapered for ease of installation and the center hinge pin(s) shall be l/4-inch shorter than the top and bottom pins. Doors shall not exceed 18 inches in width, unless hinged in the center so that when open they shall not extend more than 18 inches from the sides or ends of the unit and shall be readily removable by one man, Doors and mating parts shall be constructed to prevent inadvertent scraping of metal parts which might produce sparks.

3.4.1.1.4 Toe clearance - Toe clearance shall be provided at the operator’s position on the test equipment. This clearance shall be a minimum of 6 inches in height and 4 inches in depth.

3.4.1.1.5 Storage space - A suitable storage container or drawer shall be provided for the operating manual, handbooks, special tools, or auxiliary parts, such as adapters. This container or drawer shall be permanently attached to the test equipment in a readily accessible location.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:57Z
Check the source to verify that this is the current version before use.

3.4.1.1.6 Working surfaces - All working surfaces of the test equipment shall be spark-resistant, A suitable surface shall be provided as a desk top for recording test data. This surface shall be a minimum of 12 inches wide by 15 inches long; except for the dimensions in 3.4.3.3.1 the distance from the floor shall be specified in the detail specification. The longitudinal location of the working area shall be placed so that the operator can readily read the gages and operate the controls from this location.

3.4,1.2 Vibration - The cabinet shall be constructed so that vibration from any components shall not be transmitted through the structural members to the instrumentation to an extent which might impair readability or accuracy.

3.4.1.3 Lights - Lights shall be of the type approved by the Underwriters Laboratories for Class 1, Division 1, Group D of the National Electrical Code, Article 500-Hazardous Locations. Instrumentation other than direct-reading flow meters, shall not require integral lighting. All direct-reading flow meters shaLL include integral lighting.

3.4.1.4 Ventilation - Enclosures shall have air vent louvres near the floor level and at the top or shall be provided with an induction ventilating fan.

3.4.1.5 Air exhausting - The test equipment shall be provided with an air duct exhaust system near the item on test located well below the face of the operator. It shall provide a down draft movement of air, with suitable ducting terminating flush at the top-rear of the stand. No exhaust-air blower or fan shall be required in this system. The capacity and location of the exhaust-air intake shall be such that the operator will not be subjected to test-fluid vapor during normal operation of the equipment. A metal plate, bearing the estimated exhaust requirements in cubic feet per minute and the static pressure required to overcome the resistance in the duct work shall be attached on the top of the stand adjacent to the exhaust, The air quantities shall be sufficient to produce a capture velocity at the item being tested of not less than 50 feet per minute. Provisions shall be made for attaching additional duct lengths to the exhaust outlet.

3.4.1.6 Drip pan - A drip pan(s) shall be provided within the enclosure. The pan(s) shall be installed to catch all leakage which may drip from any component or flow down to the Inside walls of the enclosure. Drip pan(s) shall be constructed of not less than 14 gage metal. Pan(s) shall have a minimum of 1 inch slope towards the center of the unit or section. The center shall be defined as the center of the instrument panel section(s) normally facing the operator. A gutter shall be incorporated into the center of each pan which shall have a minimum of 2 inch slope from front to rear of the stand. The gutter(s) shall terminate with 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:57Z
Check the source to verify that this is the current version before use.

inch female coupling(s) for drain connection at the rear of the test equipment unit, located a minimum of 2 inches above the floor. Drain(s) shall reequipped with an approved flame-proof strainer, accessible for cleaning. Pan(s) shall be adequately reinforced to withstand, without permanent distortion, the combined weight of a 180 pound man and the components being removed or installed. Test equipment constructed in sections shall have individual drip pans installed in such a manner to prevent leakage between sections, and drip pans shall reinstalled and supported in such a manner to insure complete drainage of each section.

3.4.2 Electrical systems -

3.4.2.1 Motors - All motors shall be explosion proof and approved by Underwriters Laboratories for Class 1, Division 1, Group D of National Electrical Code, Article 500-Hazardous Locations. All motors shall be rated for continuous-duty service at the specified horsepower. Where horsepower is not specified, all motors shall be rated for continuous duty at the maximum load imposed during normal test operations.

3.4.2.1.1 AC power - AC motors shall operate on 220/440 volt, 3 phase, 60 cycle current and shall be connected for operation on 440 volts. Change over from 440 to 220 volts shall be possible with the minimum amount of additional parts. The electrical wire used shall be capable of both 220 and 440 volt operation. AC motors of 1/3 horsepower or less may be operated on 110 volt, single phase, 60 cycle current by use of a transformer, with the primary leads arranged in such a manner so that they can be selectively connected to any of the three phase legs of the power supply to allow for load balance when the unit is installed in conjunction with other components. The transformer shall be provided as a component of the test equipment, The transformer may be used to supply 110 volt power for lighting and control circuits. The current rating for continuous operation of the transformer shall be sufficient to adequately supply the 110 volt circuits without overloading, The operating temperature of the transformer shall be within the range specified in the detail specification.

3.4.2,1.2 Push button stations - Push button stations shall operate on 110 volt, single phase, 60 cycle current.

3.4,2.1.3 Remote-control components - All remote-control test equipment components, such as motor generator sets or electronic motor controls, which can be located outside a hazardous area need not be explosion proof.

3.4.2.1.4 Starters - Motor starters shall be operated by momentary contact push button stations. Starters, relays, and fuses shall be located on a remote panel or in the panel of the main control unit (MG set) of the variable speed drive where applicable, in a non-hazardous location.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:57Z
Check the source to verify that this is the current version before use.

3.4.2.1.5 Pilot lights - Green pilot-lights shall be provided on the basic test unit to indicate when each motor circuit is energized and operating properly.

3.4.2.1.6 Interlocks -Where there is more than one motor in the equipment and possible damage may be caused by stoppage of any one motor, interlocks shall be provided so that the stoppage of one motor in the circuit will not damage the equipment. If starting the motors in improper sequence can result in damage, interlocks shall be provided to insure proper sequence. Where cooling water or lubrication are required for components of the test equipment, such as vacuum pump or gear cases, the controls shall be automatic or shall be interlocked with the motor controls in a manner to prevent damage to the equipment resultant from operating error, over temperature, or over-pressure condition.

3.4.2.1.7 Belt drives - Belt driven equipment shall not be used.

3.4.2.2 Wiring - All wiring, conduit, switches, and other electrical components of the test equipment shall be explosion proof and approved by Underwriters Laboratories for Class 1, Division 1, Group D of National Electrical Code, Article 500-Hazardous Location.

3.4.2.2.1 Electrical connections - All electrical power connections to the equipment shall be made to terminal strips in a junction box located so that wires enter from the top of the test equipment. The wires shall be coded to conform with the code numbers of the wiring diagram. An additional junction box shall be provided for lighting and control circuits.

3.4.2.2.2 DC circuit - Where required to operate solenoids or other electrical components a 30 volt DC circuit shall be furnished with rheostat, ammeter, voltmeter, pilot lights, switches, and other necessary components. A red light, mounted above a warning placard, shall be provided to indicate a dangerous condition such as circuits which may inadvertently be broken or disconnected by the operator.

3.4.2 .2.2.1 DC components - Explosion-proof connectors shall be furnished with cable approved by Underwriters Laboratories, Class 1, Division 1, Group D of National Electrical Code, Article 500-Hazardous Location, to connect the component or accessory under test to the test equipment 30 volt DC power source. The 30 volt DC power supply from the test equipment shall be controlled by a standard explosion-proof switch with a red warning light. All cables supplying voltage to the item being tested shall terminate with a female connection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:57Z
Check the source to verify that this is the current version before use.

3.4.2 .2.2.2 DC power supply - The 30 volt DC power supply shall be provided as a component of the test equipment. The power supply shall include means for maintaining the operating voltage constant to within ±1/2 volt and the capability of the power supply shall be as specified in the detail specification.

3.4.2.2.3 AC circuit (400 cycle) - Where required, a 400 cycle AC circuit shall be provided which shall include all necessary instrumentation such as wiring, receptacles, controls, ammeters, voltmeters, wattmeters, etc. Type, range and capacity of instrumentation shall be approved by the procuring agency.

3.4.2 .2.3.1 AC components - Explosion-proof connectors shall be furnished with cable approved by Underwriters Laboratories for Class 1, Division 1, Group D of National Electrical Code, Article 500-Hazardous Location to connect the component or accessory under test to the test equipment 400 cycle AC power source.

3.4.2.2.4 AC power supply (400 cycle) - The 400 cycle AC power supply shall be provided as a component to the test equipment. Type and capacity of the power supply shall be as specified in the detail specification.

3.4.2.2.5 Master switch - Two palm-operated master switches, painted red, shall be provided on all test equipment exceeding 8 feet in length. Each switch shall break all circuits (electric, air, water, hydraulic, and pneumatic) simultaneously. One switch shall be located at each end of the test equipment in a readily accessible position. On test equipment up to and including 8 feet in length, a single switch shall be located at the operator's normal position.

3.4.2.3 Electrical equipment installation - All electrical equipment shall be installed in accordance with Underwriters Laboratories for Class 1, Division 1, Group D of National Electrical Code, Article 500-Hazardous Location.

3.4.2.4 Grounding - All electrical equipment in the test equipment shall be grounded in accordance with Underwriters Laboratories for Class 1, Division 1, Group D of National Electrical Code, Article 250-Grounding. A plainly marked grounding connection shall be provided for connection to the ground system.

3.4.2.4.1 Portable electrical components - Portable (auxiliary) electrical components of the test equipment shall be provided with a permanently attached (insulated) ground cable of suitable capacity for attachment to the facility

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:57Z
Check the source to verify that this is the current version before use.

grounding system. Cable shall be a minimum of 15 feet in length. A red caution tag marked “Connect to the facility grounding system prior to connecting the power supply" shall be permanently attached to the cable. A warning placard marked “Caution do not move this test unit from one location to another unless the test facility ventilating system is in operation” shall be installed on the portable (auxiliary) unit in a conspicuous location. Casters (and bearings) used on portable (auxiliary) electrical components used in hazardous locations shall be designed to prevent the possibility of static electrical discharge. Laminated phenolic tires shall be used.

3.4.3 Hydraulic system -

3.4.3.1 Reservoirs - All reservoirs shall be of corrosion resistant steel or aluminum alloy and shall be provided with readily removable clean-out plate(s). Reservoir fittings may be a part of the removable clean-out plate. Reservoirs shall be of welded construction and provided with the necessary inlets, outlets, vent, gage connections or flame arrestors when required by the detail specification; stainless steel, brass or other corrosion-resistant metal fittings shall be attached by atomic-hydrogen welding, brazing, or silver soldering. Method of attaching fittings shall be accomplished in a manner to reduce the possibility of electrolytic corrosion or erosion to a minimum. All reservoirs shall be thoroughly neutralized and cleaned for removal of weld flux, scale and other foreign matter from the interior and exterior of the reservoir. Reservoirs shall be sloped to permit complete gravity drainage of the fluid, A sump shall be provided below the inlet port of the pump.

3.4.3.1.1 Capacity - Reservoirs shall have a minimum capacity in gallons equal to or greater than 1-1/2 times the maximum flow in gallons per minute of the fluid system.

3,4.3.1.2 Baffles - Baffles, if required, shall be installed in such a manner as to prevent pressure pulsations and reduce turbulence and aeration to a minimum. Baffles shall be located with sufficient clearance at the bottom of the reservoirs to prevent an accumulation of foreign matter.

3.4.3.1.3 Reservoir pressure - The reservoir shall be designed to withstand a minimum of 2 psi internal pressure without leakage or permanent distortion (see 6.3.3(a)).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:57Z
Check the source to verify that this is the current version before use.

3.4.3.1,4 Reservoir evacuation - When required to maintain pressures less than atmospheric, the reservoir shall withstand a pressure of l/2-inch of mercury absolute, without permanent distortion or leakage, A suitable valve shall be provided below the flame arrestor to permit evacuation.

3.4.3.1.5 Reservoir drain lines - Reservoir drain lines shall be at least: 1 inch ID for reservoirs up to 100 gallons capacity, 1-1/2 inch I D for reservoirs between 100 and 150 gallons,. and 2 inch ID for reservoirs over 150 gallons capacity. Drain lines shall terminate with a standard pipe-coupling flush with the rear of the stand, extending a minimum of 8 inches above the floor, and shall be sealed with a pipe plug for protection during shipment.

3.4.3.1.6 Vents - All reservoir vents shall terminate at the top of the test equipment with a female pipe coupling provided for connection to the facility common venting system, Vent lines shall be a minimum of 1 inch ID. Vent lines shall be sealed with a pipe plug for protection during shipment, Provisions shall be made to maintain a positive pressure of not more than 2 inches of water in the reservoirs of test equipment using any volatile fluid.

3.4.3.1.7 Pressurized filling - A l-inch fill pipe provided with a shut-off valve shall be connected to the reservoir and extended to the outside rear of each unit of test equipment, for connection to the pressurized filling system. A valve shall be located close to the liquid-level gage to permit the operator to observe the gage during the filling operation, The pressurized filI-valve and liquid-level gage may be located at the rear or end of the stand. Pressurized filling shall be required only for reservoirs designed to utilize test fluids with a flash point below 170° F.

3,4.3.1.8 Line termination - All drain lines, fill lines (except manual fill line), heating, and cooling lines shall terminate flush at the outside rear of the cabinet and shall be provided with standard female couplings. Drain lines shall be provided with a shut-off valve located inside the cabinet. All lines shall be sealed with a pipe plug for protection during shipment.

3.4.3.1.9 Location of service connections - Service couplings shall be located in horizontal or diagonal relative positions with a minimum of 6 inches between each coupling to facilitate any interconnection to the facilities services.

Source: https://assist.dla.mil -- Downloaded: 202640-19T09:57Z
Check the source to verify that this is the current version before use.

3.4.3.1.10 Manual-fill connections - A fill pipe for manual filling of reservoirs shall be located within the cabinet in an easily accessible location for manual filling. The pipe shall be a minimum of 1-1/2 inch ID and shall be equipped with a tank-fill connection (including a strainer and a sealed cap) approved by the Underwriters Laboratory.

3.4.3.1.11 Strainer - A suitable strainer shall be provided between the reservoir and low-pressure centrifugal boost-pump inlet which shall be easily accessible for removal and cleaning.

3.4.3.1.12 Porting - The return port shall be located as remotely as possible from the suction port and shall be designed in such a manner that the return flow is directed away from the suction port. The return fluid shall be introduced at least 1 inch below the minimum operating fluid level.

3.4.3.2 Reservoir gages - Liquid level gages shall be provided on all reservoirs.

3.4,3.2.1 Types of reservoir gages - Hydrostatic or mechanical-type gages shall be used. The gage shall indicate the liquid level in gallons or at the 1/4, 1/2, 3/4, and full-level positions. The gage shall be properly marked to prevent overfilling of the reservoir and shall be accurate to within ±2 percent. The gage shall be marked by a red line to indicate the minimum operating level of the fluid in the reservoir.

3.4.3.2.2 Location of reservoir gages - Reservoir gages shall be located in an accessible location and shall be readable from the pressurized-fill-valve location.

3.4.3.3 Sump drain - Sump drains and working areas which collect test fluid during test shall be equipped with an approved flame arrestor and shall be piped to the outside of the test equipment. The drain line shall terminate with a pipe coupling located a minimum of 16 inches above the floor. Sump drains shall be a minimum of 4 feet in length, 18 inches in width, and not less than 26 inches nor more than 30 inches above the floor.

3.4.3.3.1 Working area - Table-top working areas shall be provided at each end of the sump drain, a minimum of 18 inches in length by 18 inches in width, and shall be a minimum of 30 inches and a maximum of 34 inches above

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:57Z
Check the source to verify that this is the current version before use.

the floor, Test equipment constructed in more than one section shall have individual working areas integral with each section. The working area and sump drain combination shall be so designed and located to facilitate installation, removal, and visuaI inspection of accessories undergoing test.

3.4.3.3.2 Waste reservoir - The waste reservoir shall not be part of the test equipment. When a waste reservoir is required it shall be located within the enclosure, shall have a minimum of 5 gallons capacity, and shall be located directly below the sump and above the waste reservoir outlet on the rear of the panel. The waste reservoir outlet shall be located a minimum of 16 inches above the floor.

3.4.3.4. Filter assemblies - All filters assemblies shall be of a size and type specified in the detail specification. The filter assembly shall be capable of handling flows at least 100 percent in excess of the maximum rated flow capacity of the test equipment without exceeding the specified allowable pressure drop.

3.4.3.4.1 Filter assembly location - The filters shall be located, when practicable, above the level of the fluid in the main supply reservoir and shall be readily accessible for servicing, Filters shall not be installed in the boost-pump inlet nor submerged in the supply reservoir.

3.4.3.4.2 Air bleed (filter assembly) - Filter assemblies shall be self purging, or provided with a l/8-inch valve connected to tubing to divert the purged fluid directly to the main reservoir.

3.4.3.4.3 Filter element replacement indicator - Red indicator lights shall be mounted on the test equipment panel to indicate when filter elements require replacement. A separate indicator light shall be used for each filter assembly. The light(s) shall indicate at the differential pressure specified in the detail specification.

3.4.3.5 Plumbing - All lines, valves, and fittings shall be designed of sufficient size to allow operation under maximum flow conditions without experiencing excessive pressure drop throughout the system. Fluid velocities in pressure lines and valves shall not exceed 15 feet per second, Fluid velocities in return and suction lines shall not exceed 5 feet per second.

3.4.3.5.1 Fluid lines and fittings - Flexible hoses, and fittings in accordance with MIL-F-5509 or MIL-F-18280 shall be used whenever possible. The use of hose clamps as fluid line couplings shall be prohibited. All valves, and fittings shall be suitable for the pressure and service to be encountered and shall be designed to withstand 150 percent of maximum operating pressure

Source: https://assist.dla.mil -- Downloaded: 2026l0-19T09:57Z
Check the source to verify that this is the current version before use.

without leakage or permanent distortion. Components such as valves and pumps may use flange connections with pipe thread outlets.

3. 4.3.5.1.1 Flexible lines - Flexible lines shall be designed to withstand 150 percent of maximum operating pressure without leakage or permanent distortion. To insure electrical continuity all flexible hose carrying test fluids shall be of the flexible steel-braid type with flared or flareless type coupling fittings installed on each end of the hose. All flexible hose shall be tested for electrical continuity. The maximum resistance of the steel braid and the hose couplings shall not exceed 100 ohms (see 6.3. 3(b)).

3. 4.3.5.1.2 Rigid lines - Rigid lines shall be of corrosion resistant steel or aluminum alloy, Rigid lines shall be designed to withstand 200 percent of maximum operating pressure without leakage or permanent distortion. The use of commercial black iron pipe or fittings shall be prohibited. The use of rigid lines shall be limited to a minimum. All pipe threads shall be in accordance with MIL-P-7105 (ANPT). The use of any material (other than thread lubricant) for sealing threaded joints shall be prohibited.

3.4.3.5.2 Panel holes - All holes through vertical panels used for installation of valves and fittings, or through horizontal surfaces normally described as the working table or area, shall be of standard size and type to permit interchange of valves and fittings from l/8-inch to l-inch size, without rework of the panel. Interchangeable flanged adapters shall be fitted to the panel holes and shall provide a rigid mounting of sufficient strength to hold the valve or fitting without additional support. Bezel plates or other means shall be provided for each panel-mounted valve or fitting which will seal the opening as effectively as practicable. Plates shall be readily removable by one man,

3.4.3.5.2.1 Auxiliary supports - All fittings, lines, tubing, and other components attached to the panel that, due to weight and size, might be damaged in shipment shall be provided with auxiliary supports or mounting brackets to minimize the possibility y of damage to the panel and panel mountings.

3.4.3.5.2.2 Rigid line mounting - Rigid lines shall not be attached to the panel mounted valves or fittings when sizes and pressures permit the use of flexible hose.

3.4.3.5.2.3 Adapters - All valves, bulkheads, adapters, and other panel mounted componets shall be mounted by screws into tapped holes or into threaded inserts.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:57Z
Check the source to verify that this is the current version before use.

3.4.3.5.3 Bulkhead ports - Bulkhead ports shall be in accordance with AND 10050.

3.4.3.6 Low pressure boost pumps - Low pressure centrifugal boost pumps shall be of a type using a carbon shaft seal rotating against steel, preferably with a steel surface stationary and the carbon seal rotating with the shaft. Stuffing-box type shaft seals shall not be used.

3.4.3.7 Fluid temperature control system - Heating and cooling of the test fluid shall be accomplished with a heat exchanger using hot or cold water where the required test fluid temperature is between 80" F and 120°F. Heat exchangers shall have a purge line installed to permit the release of air trapped in the cooling medium or in the test fluid The heat exchanger shall be designed for use with fresh water,

3.4.3.7.1 Mechanical refrigeration - When the desired test fluid temperature is less than 80°F, mechanical refrigeration equipment shall be supplied as a part of test equipment.

3.4.3.7.1.1 Refrigeration unit - The refrigeration unit shall be hermetically sealed (by fusion) and shall include a terminal box in accordance with Underwriters Laboratories, Class 1, Division 1, Group D of National Electrical Code, Article 500-Hazardous Locations to the electrical connections.

3.4.3.7.1.2 Unit capacity - The refrigeration unit shall be of sufficient capacity to permit continuous operation of the test stand at maximum output and shall not require more than 15 minutes to stablize the test fluid at the desired temperature under normal operating conditions.

3.4.3.7.2 Heat exchanger -

3.4.3.7.2.1 Cooling - The heat exchanger shall provide the required cooling when operating at an ambient temperature of 100°F using cooling water at 7 with an operating pressure of 40 to 70 psi. Heat exchangers shall withstand a maximum operating pressure of 100 psi without leakage or permanent distortion (see 6.3. 3(c)).

3.4.3.7.2.2 Heating to 180°F - The heat exchanger shall provide heating to the specified temperature when operating at an ambient temperature of 60°F and using water at 180°F or steam not exceeding 5 psi as the heating media (see 6.3. 3(d)).

Source: https://assist.dla.mil -- Downloaded: 202640-19T09:57Z
Check the source to verify that this is the current version before use.

3.4.3.7.2.2.1 Warm-up time (to 180°F) - Initial warm-up of the fluid from 60°F to operating temperature shall not require in excess of 15 minutes (see 6.3.3(e)).

3.4.3.7.2.3 Heating above 180°F - Temperatures over 180°F shall be obtained by the use of steam, The heat exchanger shall provide heating to the specified temperature when operating at an ambient temperature of 60" I?. All components of the steam system, including steam pressure regulator and condensate trap shall be provided with the unit (see 6.3. 3(d)).

3.4.3.7.2.3.lWarm-up time (above 180°F) - Initial warm-up of the fluid from 60°F to operating temperature shall not require in excess of 15 minutes (see 6.3, 3(e)).

3.4.3.7.2.4 Flow (heat exchanger) - Flow of hot or cold water shall not exceed 0.5 gallon per minute per horsepower input to the fluid system (see 6, 3. 3(f)).

3.4.3.8 Overpressure protection - Relief valves shall be provided on the discharge side of all pumps that are capable of damaging the test equipment or the item under test.

3.4.3.8.1 Relief valves - High-pressure relief valves shall be pilot operated and shall reseat at 85 percent of pressure setting. Differential pressure operating range of relief valves shall not exceed 15 percent of setting, and shall not chatter. Relief valves installed in metered lines shall indicate any leakage by means of a visible drip indicator located on the front panel (see 6.3.3(g)).

3.4.3.9 Accumulators - Accumulators shall be of the piston type in accordance with MIL-A-5498. Means shall be provided for relief of accumulator gas and hydraulic pressure.

3.4.3.9.1 Surge tanks - Surge tanks shall not permit the gas charge to come in physical contact with the hydraulic fluid.

3.4.3.9.2 Charge gas - The charge gas for accumulators and surge tanks shall be nitrogen or inert gas. A placard or tag at filler point shall indicate the type of inert gas to be used.

3.4.3.9.3 Accumulator pressure indication - A pressure gage shall be provided to indicate accumulator gas charge-pressure.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:57Z
Check the source to verify that this is the current version before use.

3.4.3.9.4 Recharging the accumulator - An air valve in accordance with MS28889 shall be provided and shall be installed in an easily accessible location for recharging the accumulator.

3.4.4 Instruments -

3.4.4.1 Flowmeters (fuel) - Flowmeters shall be as specified in the detail specification,

3.4.4.1.1 Flowmeter graduations - Meter tubes and scales of variable-area glass tube (direct reading) meters shall have logarithmic characteristics to provide uniform percentage readability at all points on the scale, and shall be calibrated in pounds per hour with at least 0.5 percent graduations. The type fluid to be used shall be as specified in the detail specification. One half of the percent of instantaneous flow at any point shall not be less than 0.625 inch. Float position readings to the nearest scale graduation shall not exceed 0.5 percent of the indicated flow. The lighting for the direct reading flowmeter shall be behind the glass indicating tube(s) with a translucent material in between. The illumination shall be evenly distributed over the translucent surface. The scales shall be of translucent material. If the flowmeter is to be calibrated for MIL-F-7024 fluids, the scale for Type I fluid shall be white and the scale for Type 2 fluid shall be green. If calibration for both Types I and 2 fluids are required on the same test equipment, the white scale shall be on the right-hand side of the tube and the green scale on the left-hand side of the tube. All other fluid scales shall be white.

3.4.4.1.2 Accuracy - The flowmeters shallhave an instantaneous accuracy within ± 0. percent when utilizing a fluid within the specified density, viscosity, and temperature limits (see 6.3.3(h)).

3.4.4.1.3 Line pressure - Flowmeters shall be designed for a line pressure of 50 psi, The meter shall withstand a leak test of 150 percent of the maximum operating pressure (see 6.3.3(i)).

3.4.4.1.4 Pressure drop - Maximum pressure drop through the flow meter or through a series of flow meters shall not exceed 8 psi. A back pressure valve may be used for the suppression of vapor release and bubbles within the instrument (see 6.3.3(j)).

3.4.4.1.5 Vent valves - The flowmeter shall have provisions for attaching vent valves to each tube which discharge through visible drip indicators that return to the main fluid reservoir. The vent valves, indicators, and return lines need not be component parts of the meter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:57Z
Check the source to verify that this is the current version before use.

3.4.4.1.6 Blow-out plugs - Flowmeters shall be protected by blow-out plugs connected so that any leakage past the blow-out plug will be indicated at a sight flow indicator located on the front panel and in a line returning by-passed fluid to the main reservoir. Blow-out plugs shall be of the diaphragm type, utilizing steel, brass, or synthetic diaphragms. Plugs shall be installed in an easily accessible location and shall be designed for quick removal and installation of the blow-out diaphragms, A strainer shall be provided in the down stream side of the blow-out plug assembly to prevent the ruptured blow-out plug from contaminating the system.

3.4.4.1.7 Single tube meters - Single tube meters shall be constructed to permit cleaning without removal of the tubes and designed so that the meters may be readily removed as a unit. The meter shall not require recalibration after cleaning and reassembly, An opening shall be provided at the top of the test equipment to permit access for cleaning the meter tubes.

3.4.4.1.8 Multiple-tube meters - Multiple-tube meters shall be assembled into a single unit including all necessary interconnecting pipe and valves.

3.4.4.1.9 Float guides - Floats utilized in flowmeter tubes shall be guided by the side wall of the tube. Construction of float guides must be such as to permit removal of float for cleaning, without disassembly of the meter.

3.4.4.1.10 Calibration ports - Calibration ports shall be provided at the inlet and outlet of the flowmeter bank, The ports shall be provided with suitable bleeds and shall be at least 5 tube diameters distant from the inlet and outlet of the bank, Ports shall be readily accessible.

3,4.4.1.11 Remote-indicating flowmeters - Remote-indicating flowmeters may be used in lieu of the variable-area glass-tube type. Accuracy shall be within $\pm 0.$ percent of instantaneous reading, and pressure drop through the meter shall not exceed 8 psi.

3.4.4.2 Flowmeters (hydraulic and lubricating oil) - Hydraulic and lubricating oil flowmeters shall be of the same type specified for fuel, (see 3.4.4.1), except that the scales shall of a yellow translucent material. Hydraulic and lubricating oil flowmeters shall be calibrated in gallons per minute and shall have an accuracy of $\pm 0.$ percent of instantaneous reading (see 6.3.3(k)).

3.4.4.3 Totalizing flowmeters - Totalizing flowmeters, where required, shall be as specified in the detail specification.

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:57Z
Check the source to verify that this is the current version before use.

3.4.4.4 Motor and pump tachometer - The range, degree of accuracy, size, and type of tachometer shall be as specified in the detail specification. Electrical components shall be easily removable by plug-in type construction and shall be contained in one chassis which can be mounted in a non-hazardous area up to 500 feet from the hazardous test area so that no electronic circuity other than the indicator need be located in the hazardous area,

3.4.4.4.1 Calibrating - Provisions shall be made for checking the calibration of tachometers, other than electronic, without removal from the stand either by checking the shaft speed with the hand tachometer or by incorporation of a common junction box with disconnect plugs for checking with auxiliary test equipment.

3.4.4.5 Timers - Where precision timers are required, the timer shall be self contained and shall not require an external source of frequency control.

3.4.4.6 Temperature indicators - Temperature indicators shall be of a type and range as specified in the detail specification,

3.4.4.7 Pressure gages - All panel mounted pressure gages shall be recess-type provided with mounting flange at the front of gage. The mounting flange (and dial) shall not extend more than 5/8-inch beyond the external surface of the panel. All gages shall incorporate a safety blow-out device to prevent blow-out through the face of the gage. The safety blow-out device shall automatically relieve the gage case-pressure should it reach l/2-psi due to gage failure. The pointer shall be adjustable from the face of the gage. Stainless steel or nylon movements shall be required on all gages. Gages subjected to continuous surges and extreme pressure fluctuations shall be protected by a hydraulic damper. The type, size, range, and accuracy of pressure gages shall be as specified in the detail specification. The dial pointer shall be separated from the gage movement such as to prevent rupture through the front of the gage in the event of gage failure.

3.4.4.7.1 Mounting - Gages shall be mounted utilizing screws and tapped holes to permit removal from the front of the panel by one man.

3.4.4.7.2 Connections - Gages shall be connected to the panel with flexible hose of sufficient length to permit removal and installation from the front of the panel.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:57Z
Check the source to verify that this is the current version before use.

3.4.4.7.3 Throttling - All pressure gages shall be equipped with needle valves located as close as practicable to the gage.

3.4.4.7.4 Protection - Gages subject to pressure beyond their maximum range shall be protected with automatic gage shut-off valves. Automatic gage shut-off valves shall be of positive action which will allow no leakage of fluid to the gage. A suitable in-line filter shall be provided in the circuit immediately ahead of the shut-off valve. The differential between the cut-off and cut-on point shall not exceed ±1 percent of the range of the gage,

3.4.4.7.5 Calibration taps - A calibration tap shall be included between each gage and the throttling valve to facilitate the calibration of the pressure gage from the front of the test equipment.

3.4.4.7.6 Air bleed (pressure gages] - Where maximum accuracy is required in low pressure gages and where the fluid head in the lines will cause a significant error in the gage reading, provisions shall be made to purge gage lines of air and maintain a constant liquid head on the gage.

3.4.4.7.7 Drip indicator - Bleed lines on pressure gages shall be provided with visible drip indicators.

3.4.4.7.8 Taps - All pressure gages shall be provided with a tap on the panel between the gage and the gage shut-off valve to permit calibration of the gage without removal from the panel.

3.4.4.8 Measuring section - Where it is necessary to measure inlet or discharge pressure in the lines connected to the accessory, special measuring sections shall be included with the equipment. Measuring sections shall be provided only on stands designed for testing fuel controls, fuel nozzles, and fuel regulators.

3.4.4.8.1 Design of measuring sections - Each section shall consist of a length of straight stainless steel tubing with an inside diameter the same as the standard fitting installed in the accessory pressure tap. A gage connection shall be installed in this tubing preceded by 5 diameters length of tubing and followed by 5 diameters length of tubing. The tubing and connection shall be free of any restrictions which could affect the accuracy of the indicated pressure.

3.4.4.8.2 Couplings - Quick-connect couplings providing unrestricted streamlined flow shall be used between the measuring section and the accessory.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:57Z
Check the source to verify that this is the current version before use.

3.4.4.9 Manometers - Type and range of manometers shall be as specified in the detail specification, Absolute pressure manometers used for setting Up density shall be of the density-calculating type and shall be calibrated for the densities specified in the detail specification.

3.4.4.9.1 Pressure - All manometers shall withstand 125 percent of operating pressure without leakage or permanent distortion.

3.4.4.9.2 Valves - Suitable valves (including a fluid trap) which will prevent the indicating fluid from being forced out of the manometer shall be installed at the top of the manometer column and in the well connection.

3.4.4.10 Specific gravity testers - All fuel component test equipment shall be equipped with a suitable specific gravity tester having an accuracy of ±0. percent. A thermometer shall be provided with the tester or shall be installed in the fluid stream to the tester.

3.4.4.10.1 Range - The hydrometer for the specific gravity tester shall be selected for range of fluid being used and shall not rotate in the well.

3.4.4.10.2 Well - The hydrometer well shall be provided with valves to allow fresh samples of the circulation fluid to be taken, tested, and returned to the tanks.

3.4.4.10.3 Flow specific gravity tester) - Flow through the hydrometer shall not be indicated on the flowmeter.

3.4.4.11 Torque measuring equipment - Where it is necessary to measure the torque required to drive the item on test, a direct-indicating torque meter shall be provided, Dead-weight type equipment adaptation shall also be provided on the test stand to check the calibration of the torque meter.

3.4.4.12 Viscosimeters - A viscosimeter shall be provided as a component on the test equipment. Viscosimeters shall be provided with valves to allow fresh samples of circulating fluid to be ‘taken, tested, and returned to the reservoir.

3.4.4.13 Readable accuracy - The instruments shall be selected so that they are readable to the accuracy described in this and the detail specifications.

3.4.5 Controls -

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:57Z
Check the source to verify that this is the current version before use.

3.4.5.1 Temperature control - Automatic controls shall be provialed for heating and cooling the test fluid and shall maintain the set temperature within ±1°F during constant and transient operation.

3.4.5.1.1 Manual temperature control - Manual controls shall be furnished only to trim the hot and cold water supply for better flow control and for use as shut-off valves.

3.4.5.1.2 Span - The span of automatic temperature-control adjustment shall be as specified in the detailed specification,

3.4.5.1.3 Operation (control) - The automatic temperature control shall operate with water from + 40”F to + 200°F and steam from + 212°F to +

3.4.5.2 Manual operating controls - All controls shall be selected and arranged to provide a maximum of convenience in operating ease, to reduce test time and operator fatigue,

3.4.5.2.1 Location - All controls shall be located so as to permit all phases of the test to be accomplished by one operator from the front of the test equipment.

3.4.5.2.2 Remote operation - Control valves larger than 3/4-inch port size, shall be remotely operated by air with regulator valve located on the front panel. The control/valve shall be designed for continuous operation with compressed air at 150 psi. The controls shall be such that in the event of electrical or air failure the valve will be returned to “rest” or “safe” position automatically (see 6.3.3(1)).

3.4.5.2.3 Shut-off and control valves - The handle motion of the valve shall not exceed 90 degrees of rotation between extreme positions. Handle loads shall be minimized by the use of ball thrust bearings or by balanced pressures. Detents or nameplates shall be provided to identify correct handle positions. The flow passage diameters in the valves shall be equal to or greater than the internal diameter of the piping to which they are connected. The valves shall so operate that no undue hydraulic shocks or surges shall result from their use to change flow volume or direction, The maximum allowable torque required to operate manual-control valves shall not exceed 40 pound-inches. Push-pull types of valves may be used provided they meet applicable requirements (see 6.3.3(m)).

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:57Z
Check the source to verify that this is the current version before use.

3.4.5.2.3.1 Hystersis - Valves shall not creep from selected position and shall be free from hysteresis,

3.4.5.2.4 Needle valves - Needle valves shall be panel-mounted. AH needle valves used shall show the manufacturer's operating pressure rating stamped on the valve body, as well as an arrow showing flow direction. Needle and packing shall be readily accessible from the front of the stand. The valve knob or handle shall be installed on a square or milled flat-section of the valve stem. Valves shall be of a type that can be disassembled for inspection, repair or replacement of parts.

3.4.5.3 Automatic controls - Where it is necessary to control pressure or flow to a constant value, automatic controls shall be used. A slight amount of adjustment of the control will be permitted when it is necessary to control within close limits over a wide range of operating conditions. Limits and accuracy shall be as specified in the detail specification.

3.4.5.4 Automatic shut-off valves - Automatic shut-off valves shall be inter-locked to operate automatically with the test equipment emergency-disconnect controls. Automatic shut-off valves other than those in the flowmeter system shall be pneumatic or solenoid actuated valves. In the event of test stand power, water, or air failure automatic shut-off valves shall fail in the safe position.

3.4.6 Test fixtures - Test fixtures for test equipment designed to test only one type of accessory shall be furnished as component parts of the equipment.

3.4.6.1 Multiple purpose - Test fixtures for test equipment designed to test various types of accessories shall not be included with the basic test equipment, with the exception of the base mounting fixture which shall be furnished.

3.4.6.2 Location - Mounting pads and brackets for test fixtures shall be conveniently located in relation to the related instruments and controls. Test fixtures shall not interfere with the instrument and control adjustments that are required during calibration.

3.4.7 Drive pad - All drive pads for testing driven components shall be designed so that the mounting pad and spline adapter will be fully removable and interchangeable with other sizes ofsplines and pads. The pads and splines shall bc easily removed and installed from the front of the stand by one man. Pads and splines shall be in accordance with AND 20001 and AND 20002, as applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:57Z
Check the source to verify that this is the current version before use.

3.4.7.1 Quick-change clamps - All drive pads shall be equipped with quick- change clamps and shall have a dowel pin for each hole in the accessory mounting flange.

3.4.7.2 Dowel pins - Diameter, length, and material used for dowel pins shall be as specified in the detail specification.

3.4.7.3 Clamps - The design, and configuration of quick-change clamps required, shall be as specified in the detail specification,

3.4.8 Variable-speed drives - Minimum and maximum operating speeds, horsepower values for the various operating speeds, type of drive, (reversing or non-reversing) and type of speed regulation (non-automatic or automatic) of variable-speed drives shall be as specified in the detail specification. The speed control shall provide infinitely variable adjustment throughout the speed range and shall be free of hysteresis, creep and backlash.

3.4.8.1 Regulation - Speed regulation shall be +0.5 percent, and is defined as follows (see 6.3.3(n)):

3.4.8.2 Response - The maximum deviation from the set speed, upon sudden variation of load shall not exceed 1 percent of the set speed. This deviation shall occur in the direction of load change only, with no subsequent overshoot. All transients resulting from sudden load variations or sudden changes in the set speed shall be effectively damped out in less than O. 1 second (see 6.3.3(n)).

3.4.9 Motor sets - Each driving and driven member shall be mounted on a common base. All motor sets shall be shock mounted to reduce vibration and noise to a minimum. The shock mounting shall be designed to effectively insulate the assembly from the test unit so as not to transmit perceptible vibration to the installed instruments.

3.4.10 Vacuum system - Where vacuum is necessary to meet test requirements, a complete self-contained vacuum system shall be furnished. This system shall consist of an electric motor and vacuum pump unit of sufficient capacity and stability to maintain the test conditions.

3.4.11 Exhaust - Vacuum pumps used for evacuating fuel or oil tanks and systems shall have an independent exhaust line extending to the rear of the test equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:57Z
Check the source to verify that this is the current version before use.

3.4.12 Compressed air regulating system - A compressed air regulating system shall be provided and shall be of adequate size and stability for continuous operation. System components shall consist of a 0-200 psi gage to indicate pressure at source, a filter and trap, a regulator (200 psi rated), and a suitable gage to indicate regulated pressure.

3. 4.12.1 High pressure air system - The high pressure air system shall be as specified in the detail specification.

3.4.13 Lubrication - AU components requiring periodic lubrication shall be provided with fittings in accordance with MIL-F-3541. Lubrication fittings shall be installed in easily accessible locations and a placard shall be provided adjacent to each fitting indicating the type of lubricant to be used. Components installed within the enclosure at any inaccessible locations shall be connected by tubing (rigid or flexible) to an adjacent location outside the structure and above the level of the component.

3.4.14 Noise level - The noise level of any particular operating section or the combined sections together shall not exceed 90 decibels.

3.5 Performance - The test equipment shall meet the performance requirements of this specification when tested as specified under 4, 6.

3.6 Finish - All exposed parts and surfaces of the test equipment and it components shall be finished as specified herein.

3.6.1 Corrosion-resistant metal parts - All protected metal parts and surfaces that are constructed of corrosion-resistant material and on which a protective coating would be detrimental to proper functioning, shall not be given protective coatings. All exposed metal parts and surfaces, and all protected metal parts on which a protective coating will not be detrimental to proper functioning shall be treated to satisfy the requirements of this specification.

3.6.1.1 Ferrous materials - Ferrous surfaces shall be cleaned and given protective treatment by either of the following methods:

a. Cleaning treatment in accordance with TT-C-490, Type I and primer coating in accordance with TT-P-636.

b. Cleaning treatment in accordance with TT-C-490, Method I or II and primer coating in accordance with MIL-P-26915.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:57Z
Check the source to verify that this is the current version before use.

All ferrous items such as screws, bolts, nuts, and washers shall be cadmium plated in accordance QQ-P-416, Type II or III, as applicable, or zinc plated in accordance with QQ-Z-325.

3.6. 1.2 Aluminum and aluminum alloy - Aluminum and aluminum alloy parts and surfaces shall be cleaned in accordance with MIL-S-5002, A corrosion-preventive chemical film conforming to MIL-C-5541, followed by a primer coating conforming to MIL-P-8585, shall then be applied,

3.6.1.3 Magnesium alloy - Magnesium alloy parts shall not be used unless specifically approved for each individual application by the procuring activity. When magnesium alloy parts are used they shall be cleaned and chemically treated in accordance with MIL-M-3171. Two primer coatings in accordance with MIL-P-8585 shall then be applied,

3.6.1.4 Other surfaces - Other types of surfaces shall be cleaned and chemically treated in accordance with MIL-T-704. A primer coating conforming to MIL-P-8585 shall then be applied,

3.6.1.5 Finish coatings - Exterior surfaces of the test equipment shall be finished with two coats of enamel as specified in the detail specification. The external color shall be semi-gloss light green number 24441 conforming to FED-STD-595. Interior surfaces shall be finished with two coats of enamel as specified in the detail specification. The internal color shall be either glossy white number 17875 or light gray number 16492 conforming to FED-STD-595.

3.7 Identification of product - The test equipment shall be marked for identification in accordance with MIL-STD-130 and the following subparagraphs.

3.7.1 Vendor's nameplate - The vendor's standard nameplate may also be installed in addition to the marking specified in the detail specification.

3.7.2 Placards - A placard shall be permanently mounted on the panel listing all valves, gages, outlets, connections, and controls identified by name and number. The same numbers shall be shown on the flow diagram and other data furnished with the test unit.

3.7.2.1 Service lines - Steam, water, air, and drain lines shall be marked at entrance to the equipment and shall be located at the rear of the test equipment unit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:57Z
Check the source to verify that this is the current version before use.

3,7,2.2 Warning - Warning placards or other precautionary labels, as necessary, shall be so located and designed that the operator cannot fail to notice them before any controls, valves or switches are operated, Warning placards shall be red with white engraved letters and constructed from laminated phenolic sheet,

3.7,2.3 Mounting adapters - All mounting adapters and drive couplings shall be identified by a separate manufacturer's part number stamped on each item.

3,7.2.4 Rotation - Directions of rotation on all components shall be properly identified with arrows applied by engraving or other permanent method of marking,

3.8 Workmanship - The test equipment, including all parts and accessories, shall be constructed and finished such as to assure ability of all equipment and components to meet their performance requirements under all applicable service conditions. Particular attention shall be given to: neatness of wiring; proper installation of tubing, hose, and fittings; thoroughness of welding, brazing, drilling, forming, riveting, and painting; alignment of parts; tightness of screws and bolts; and freedom from defects, burrs, and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4, 1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein, Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection - The inspection of the test equipment shall be classified as follows:

(a) Preproduction inspection

(b) Quality conformance inspection

4.3 Preproduction inspection - Preproduction inspection shall consist of all the inspections of this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:57Z
Check the source to verify that this is the current version before use.

4.3.1 Sampling for preproduction inspection- Unless otherwise specified, as soon as practicable after award of a contract or order, the contractor shall furnish a preproduction sample for inspection to determine conformance with this specification. The preproduction sample shall consist of one complete hazardous location test equipment system on contract or order. The sample shall be identified with the manufacturer's part number and contract number and shall be inspected as specified herein and in accordance with the contract or order (see 6. 2).

4.4 Quality conformance inspection - The quality conformance inspection shall consist of individual inspection only,

4.4.1 Individual inspection - Each test equipment system on the contract or order shall be subjected to the following inspection in the order listed. Any test equipment system containing a defect or failing to pass any of these tests shall be rejected.

(a) Examination (4. 6.1)

(b) Operation (4. 6.2)

(c) Correlation (4, 6.3)

4.5 Test conditions - Unless otherwise specified, tests shall be conducted at an ambient temperature of 77 ± 18° F and at an ambient barometric pressure of 28 to 32 inches of mercury. If tests are conducted at different ambient conditions, appropriate correction factors shall be applied to the flow and pressure readings.

4. 5.1 Test fluids - Test stands shall be tested using the type fluid as specified on the f low meter calibration scales.

4.6 Inspections methods -

4.6.1 Examination - The test equipment system shall be thoroughly examined to determine conformance with this specification and applicable drawings with respect to all the requirements not covered by tests.

4.6. 1.1 The test equipment system shall be examined to ascertain that the preparation for delivery conforms to this specification.

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:57Z
Check the source to verify that this is the current version before use.

4.6.2 Operation of test equipment - All test equipment units shall be operated as specified in the detail specification at full rated capacity for a minimum of two hours and shall maintain proper test fluid temperature, speed and speed controls with no evidence of overloading or other indications of stress or failure.

4. 6.3 Correlation - The test equipment shall be subjected to and shall meet the requirements of the correlation tests specified in the detail specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging -

5, 1.1 Level A - Unless otherwise specified each test equipment stand shall be preserved and packaged in accordance with MIL-STD-794, Level A. The packaging method shall conform to MIL-P-116, Method II, The hydraulic system of the test equipment shall be preserved with a hydraulic equipment preservative compatible with the test fluid to be used in operation of the test equipment.

5. 1.2 Level C - Unless otherwise specified preservation and packaging shall conform to MIL-STD-794, Level C. The hydraulic system of the test equipment shall be preserved with a hydraulic equipment preservative compatible with the test fluid to be used in operation of the test ‘equipment.

5.2 Packing - Unless otherwise specified patting shall conform to MIL-STD-794, Level A, B, or C as specified in the contract or order.

5.3 Marking - Unless otherwise specified marking shall conform to MIL-STD-129. The nomenclature of the equipment marked on the container shall be as indicated in the detail specification. The packages, container, and shipping documents shall include the complete SCC code number.

6. NOTES

6.1 Intended use - The test equipment is used to test various aircraft and missile components used in engine, hydraulic, fuel, and pneumatic systems.

6.2 Information for contracting officer - Contracts or orders should specify the following:

(a) Title, number, and date of this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:57Z
Check the source to verify that this is the current version before use.

(b) Selection of applicable levels of preservation, packaging, and packing required.

(c) Items of data required (see 6. 3).

6.2.1 Contracts or orders should specify the following provisions for preproduction inspection.

6. 2.1.1 Whether preproduction inspection is required. When a contractor is in continuous production of the items from contract to contract, consideration should be given to waive the preproduction inspections. If preproduction inspection is required, indicate:

(a) where the preproduction inspection is to be conducted (at the contractor's plant or Government or commercial laboratory).

(b) that the approval of preproduction samples or the waiving of the preproduction inspection shall not relieve the contractor of his obligation to fulfill all other requirements of the specification and contract.

6.3 Data - For the information of Contractors and Contracting Officers , any of the data specified in (a) subparagraphs below, (b) applicable documents listed in Section 2 of this specification, or (c) referenced lower-tier documents need not be prepared for the Government and shall not be furnished to the Government unless specified in the contract or order, The data to be furnished shall be listed on DD Form 1423 (Contractor Data Requirements List), which shall be attached to and made a part of the contract or order.

6. 3.1 Preproduction data - When preproduction samples are submitted (see 6. 2) they should be accompanied by one complete set of manufacturer's drawings, the contractor's model specification, a parts list, and a complete test report showing the results of the contractor's tests. The test report should include the following:

(a) Report of all tests, graphically presented when possible, together with a detailed statement indicating compliance or extent of noncompliance with all requirements of this specification, referring specifically to paragraph numbers. Wherever a requirement is considered to be not applicable, the report should so state.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:57Z
Check the source to verify that this is the current version before use.

(b) Diagrams of test setups.

(c) Reproducible outline and description of tests and test conditions. Where tests specified in this specification are not considered applicable, the reason, and the substituted test should be clearly described.

(d) Copies of test log sheets.

(e) Photographs when available.

6.3.2 Drawings and data list - Drawings, engineering and associated lists of the test equipment shall be in accordance with MIL-D-70327 (Drawings, Engineering and Associated Lists).

6.3.3 Verification data - A report shall be submitted verifying that components have met the following design, test, calibration, and performance requirements specified in Section 3:

(a) Reservoir internal pressure test (see 3.4.3.1.3)

(b) Braided hose and coupling electrical resistance (see 3.4.3.5.1.1)

(c) Heat exchanger maximum operating pressure test (see 3.4.3.7.2.1)

(d) Warm-up test (see 3.4.3.7.2.2 or 3.4.3.7.2.3)

(e) Warm-up time (see 3.4.3.7.2.2.1 or 3.4.3.7.2.3.1)

(f) Heat exchanger flow (see 3.4.3.7.2.3)

(g) Relief valves (see 3.4.3.8.1)

(h) Fuel flowmeter accuracy (see 3.4.4.1.2)

(i) Flowmeter leak test (see 3.4.4.1.3)

(j) Flowmeter pressure drop (see 3.4.4.1.4)

(k) Hydraulic and lubricating oil flowmeter accuracy (see 3.4.4.2)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:57Z
Check the source to verify that this is the current version before use.

(1) Control valve airpressure test (see 3.4.5.2.2)

(m) Shut-off valve operating torque test (3.4.5.2.3)

(n) Variable speed regulation and response (3.4.8.1 and 3.4.8.2)

6. 3.4 Installation data - Installation data shall be provided not less than 30 days prior to the preproduction tests. This data shall consist of test stand characteristics such as weight, volumetric dimensions, special foundation requirements, utility and services requirements (including ventilation, air, steam, water, electricity, drains, etc.). One copy of commercial installation data shall be packed with each test equipment. Three additional copies shall be shipped in accordance with instructions to be furnished to the vendor.

6. 3.5 Schematics - Electrical, pneumatic, lubrication, hydraulic, and fuel systems schematics of the test equipment shall be permanent y affixed to the equipment in a readily available position for reference by the operator, These schematics shall be clear and easily interpreted by using standard codes. Schematics shall be a minimum of 8 inches by 10 inches. Test equipment built in sections shall have sectional schematics in addition to the composite schematic.

6. 3.6 Handbooks - A handbook in accordance with MIL-H-7960 (Handbook, Operation and Service Instructions (Test Equipment and Support Equipment)) shall be provided. A tabulation of the test equipment characteristics such as weight, floor space, utilities and facilities requirements shall be included as well as photographs illustrating salient features of the test stand. Three copies of the handbook shall be supplied and shipped with each test equipment. Three copies of the handbook shall also be furnished,

6.3.7 Commercial literature - Three complete sets of commercial literature or handbooks covering all the commercial components of the basic test equipment, normally furnished by the manufacturer of commercial vendor items shall be packed with each test equipment in loose leaf form. The handbooks or commercial literature shall contain the manufacturer's operating and service instructions. Three complete sets shall also be furnished.

6.3.8 Data sheets - A data sheet covering the test equipment shall be prepared in the form of a reproducible copy in accordance with MIL-D-19731 (Data Sheets for Aircraft and Missile Systems Support Equipment). The data sheet shall be approved. The final draft shall be provided, The data sheet shall include the complete Standard Commodity Classification (SCC) number for the test equip-

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:57Z
Check the source to verify that this is the current version before use.

ment. This includes the 12 digit SCC code number and the 5 digit manufacturer's code number. This information can be found in the "Defense Supply Management Handbook DSMA H-3, Department of Defense Classification of Equipment for Inventory Control, June 1953" and the "Federal Supply Code for Manufacturers, July 1956 (H4-1 and H4-2"".

6. 3.9 Provisioning documentation - Provisioning documentation in conformance with MIL-M-8910 (Manuals, Technical, Illustrated Parts Breakdown, Preparation of) shall be provided for inspection at the time of preproduction tests. If satisfactory, the reproducible list and three copies shall then be forwarded.

6,4 Changes from previous issue - The extent of changes (deletions, additions, etc.) preclude the annotation of the individual changes from the previous issue of this document.

Custodians:

Army-MO
Navy-WP
Air Force-82

Preparing Activity:

Navy-WP

Review Activities:

Army-MO, MI
Navy- WP, MC
Air Force-82, 71

(DoD Project No, 4920-0009)

User Interest:

Air Force- 11

Review/user information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current Federal Supply Classification Listing of DoD Standardization Documents,

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:57Z
Check the source to verify that this is the current version before use.

INDEX

AC circuit (400 cycle) -electrical systems	3.4.2.2.3
AC power - to operate test equipment	3.4.2.1.1
AC power supply (400 cycle) - electrical systems	3.4. 2. 2.4
Accessibility and serviceability - structure	3.4.1.1.3
Accumulators - hydraulic system	3.4.3.9
Adapters - plumbing	3.4.3.5.2.3
Air exhausting - structure	3.4.1.5
Aluminum and aluminum alloy - finish	3.6.1.2
Belt drives - electrical systems	3.4.2.1.7
Commercial literature - data	6.3.7
Compressed air regulating system	3.4.12
Controls	3.4.5
Corrosion - resistant metal parts - finish	3.6.1
Data	6.3
DC circuit - electrical systems	3.4.2.2.2
DC power supply - electrical systems	3.4.2.2.2.2
Design and construction	3.4
Drawings and data lists - data	6.3.2
Drip pan - structure	3.4.1.6
Drive pad	3.4.7
Electrical equipment installation - electrical systems	3.4.2.3
Electrical systems	3.4.2
Ferrous materials - finish	3.6.1.1
Filter assemblies - hydraulic system	3.4.3.4
Finish - metals, panels, structure	3.6
Flowmeters (fuel) - instruments	3.4.4.1
Flowmeters (hydraulic and lubricating oil) - instruments	3.4.4.2
Fluid temperature control system - hydraulic system	3.4.3.7
Grounding - electrical systems	3.4.2.4
Grouping of systems, components, and cabinet	3.4.1
Handbooks - data	6.3.6
Heat exchanger - hydraulic system	3.4.3.7.2
Hydraulic system	3.4.3
Identification of product	3.7
Information for contracting officer	6.2.6.2.1, and 6.2.1.1
Installation data	6.3.4
Instruments	3.4.4
Interlocks - electrical systems	3.4.2.1.6

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:57Z
Check the source to verify that this is the current version before use.

Lights - structure	3.4.1.3
Low pressure boost pumps -hydraulic system	3.4, 3,6
Lubrication	3.4.13
Magnesium alloy -finish	3.6.1.3
Manometers - instruments	3.4.4.9
Master switch - electrical systems	3.4.2.2.5
Materials	3.3
Measuring section - instruments	3.4.4,8
Motors - electrical systems	3.4.2.1
Motor sets	3.4.9
Noise level	3.4.14
Overpressure protection - hydraulic system	3.4.3.8
Panel holes - plumbing	3.4.3.5.2
Pilot lights - electrical systems	3.4,2.1.5
Plumbing - hydraulic system	3.4.3.5
Preproduction data - data	6. 3.1
Pressure gages - instruments	3.4.4.7
Refrigeration - hydraulic system	3.4.3.7.1
Relief valves - hydraulic system	3.4,3.8.1
Remote - control components - electrical systems	3.4.2.1.3
Reservoirs - hydraulic system	3.4.3.1
Schematics - data	6.3.5
Sections - structure	3.4.1.1.1.1
Size - structure	3.4.1.1.1
Specific gravity testers - instruments	3.4.4.10
Starters - electrical systems	3.4.2.1.4
Structure	3.4.1.1
Sump drain - hydraulic system	3.4.3.3
Tachometer - motor and pump	3.4.4.4
Tank evacuation (vapor)	3.4.11
Temperature indicators - instruments	3.4.4.6
Test fixtures	3.4.6
Timers - instruments	3.4.4.5
Torque measuring equipment - instruments	3.4.4.11
Vacuum system	3.4.10
Valves - shut off, control, needle	3.4.5.2.3, 3.4.5.2.4, 3.4.5.4
Variable - speed drivers	3.4.8
Ventilation - structure	3.4.1.4
Verification data - data	6.3.3
Vibration - structure	3.4.1.2
Viscosimeters - instruments	3.4.4.12
Wiring - electrical systems	3.4.2.2

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:57Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:57Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION MIL-T-21578A, TEST EQUIPMENT; HAZARDOUS LOCATION, INSTALLED; BASIC REQUIREMENTS FOR

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:57Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
Naval Air Engineering Center
Philadelphia, Pennsylvania 19112

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

BuWeps Engineering Standardization Office (Code X)
Naval Air Engineering Center
Philadelphia, Pennsylvania 19112

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:57Z
Check the source to verify that this is the current version before use.